Invariance and selectivity to spatial variables in deep neural networks and the brain

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Abstract

Although it is generally thought that the formation of cognitive maps requires the brain to integrate self-movement (i.e., path integration), recent studies challenge this view by showing the existence of spatial selectivity in deep networks trained with vision inputs alone. Because these studies only examined neuronal selectivity to individual spatial variables (e.g., location, head direction), without characterizing their joint tuning, it remains possible that the ways in which space is encoded in the brain and in these models are substantially different. We develop an analysis framework for the joint tuning of neurons to multiple spatial variables and apply it to the rodent brain and to recently proposed vision-based deep networks. Neurons in the brain and in the models show qualitatively different selectivity and invariance to location and head direction. Our results demonstrate the importance of studying joint tuning properties in understanding neural codes.

1 Introduction

Examining the tuning property of individual neurons is a basic approach for studying the neural code [1, 2, 3, 4, 5, 6, 7]. Over the past few decades, neuroscientists have characterized the tuning properties of neurons to behavioral variables in many brain regions (e.g., [3, 4, 5, 6, 8]). In the hippocampus and related brain areas (e.g., entorhinal cortex and parahippocampus), neurons selective to various spatial variables (e.g., location, head direction, speed) have been discovered [2, 9, 3, 6, 10, 8]. Earlier studies focused mainly on the selectivity to one stimulus variable (e.g., visual orientation, location, or head direction). More recent studies also examined the selectivity of neurons to multiple stimulus dimensions individually. If a neuron is selective to multiple variables, the neuron is considered to exhibit conjunctive tuning [11] or mixed selectivity [12, 13, 14].

A large body of work has investigated how the brain constructs spatial maps of the environment. This question is relevant both for understanding how the brain supports navigation [15] and for building embodied intelligent agents [16]. It is generally believed that the brain integrates self-movement signals (i.e., velocity input and head rotation) to generate such maps [15] (the path integration hypothesis [17]). Other sensory inputs, such as visual landmarks and boundaries, are thought to recalibrate the cognitive map [18, 19]. Recurrent neural network models, either hand-crafted [20, 19, 21] or optimized for navigation tasks [22, 23, 24, 25, 26], have been developed to model place cells, head direction cells, and grid cells. Recent studies challenged this view. In deep neural network models trained to solve purely visual tasks, many network units exhibit spatial selectivity that resembles that observed in the rodent hippocampus [27, 28]. When comparing the brain and the models, these studies examined the marginal tuning of each neuron to individual stimulus variables, one at a time. Marginal tuning curves describe the joint tuning only when the joint tuning factorizes into the product of the marginal tuning curves; otherwise they can misrepresent how a neuron depends on each variable.

Here we directly analyze the *joint tuning* of individual neurons to multiple stimulus variables *simultaneously*. We first show that the tuning curve for one variable at a time can miss important information about the tuning properties. We then develop an analysis framework for the selectivity and invariance of the neuronal response when multiple stimulus variables are considered together. Applying this framework to rodent spatial navigation circuits and to recently proposed deep neural network models of these circuits yields three main findings. First, rodent place cells are selective to the animal's location and largely invariant to head direction. Conversely, head direction cells are selective to head direction and largely invariant to the animal's location. Second, units in deep neural network models trained to perform vision tasks [28, 27] exhibit entangled tuning for location and head direction, and are therefore markedly distinct from place cells and head direction cells in rodents. Third, rodent head direction cells are largely invariant to movement direction, which has implications for models of spatial navigation circuits. Using spatial navigation as an example, our results demonstrate the importance of analyzing the joint tuning of neurons in order to understand the structure of the representation in the brain and in deep neural networks.

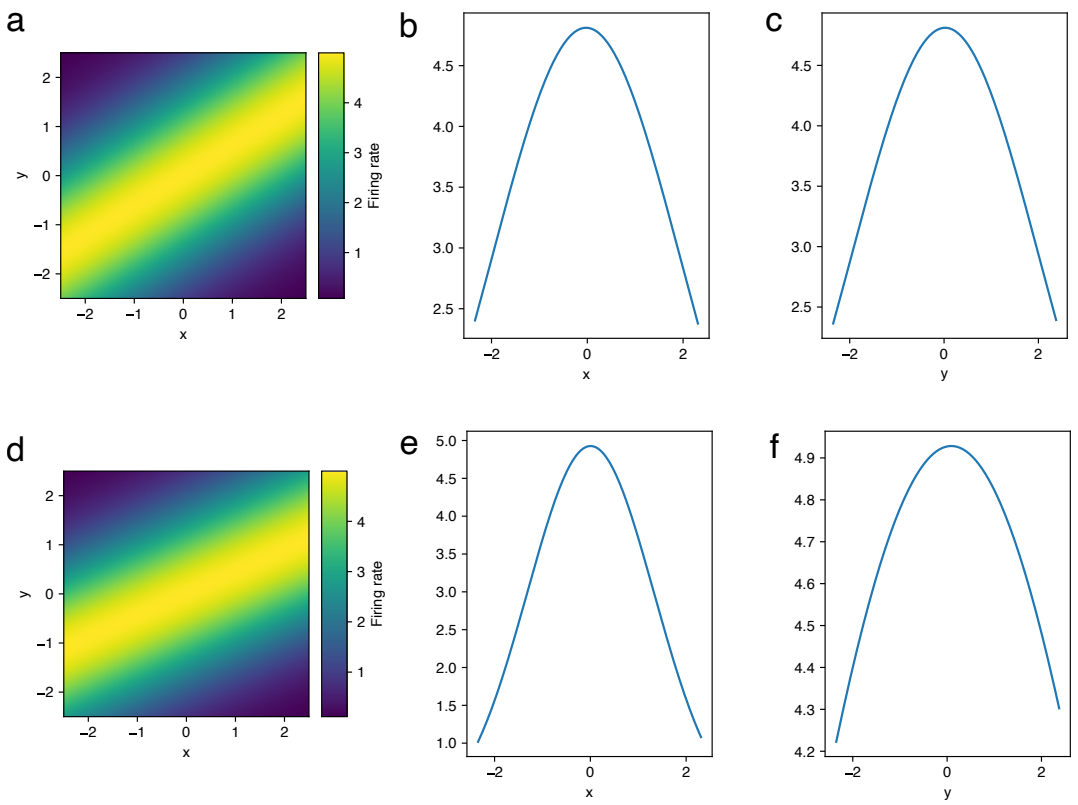


Figure 1: A generalized linear model (GLM) with separate terms for each variable does not capture the joint tuning curve. (a) The ground truth joint tuning curve for a hypothetical neuron. This neuron exhibits entangled joint tuning to variables x and y. The marginal tuning curves for x and y are both flat. (b,c) Fitting a GLM with separate terms for x and y to simulated responses from this neuron yields marginal tuning curves for both variables that are not flat, which is misleading. (d–f) Same as (a–c), for a different hypothetical neuron.

2 Results

2.1 Marginal tuning curves are insufficient to capture the joint tuning

When only one stimulus variable (denoted x, such as the head direction of the animal) is considered, the tuning property of a neuron can be described by its tuning curve $f(x)$, which quantifies how the firing rate of the neuron varies with the stimulus variable. Suppose that another stimulus variable y is also of interest (e.g., the movement direction of the animal), and that our goal is to understand whether this neuron is selective to both x and y. The standard approach is to examine the marginal tuning curves $f(x)$ and $g(y)$ separately to see whether one or both exhibit strong selectivity. If $f(x)$ is a narrow Gaussian function and $g(y)$ is flat, it is tempting to conclude that the neuron is selective to x and invariant to y. However, this analysis of the marginal tuning curves can misrepresent the joint tuning to the two variables.

Fig. 1 illustrates two such examples. The *joint tuning map* $h(x, y)$ for the first example is plotted in Fig. 1a. The marginal tuning curve $f(x)$ is tuned, while $g(y)$ is flat. Although the marginal tuning curve for y indicates a lack of selectivity, the joint tuning map shows that the response of this neuron is not invariant to y: the response depends on y through its interaction with x. In the second example (Fig. 1d),

the marginal tuning curves $f(x)$ and $g(y)$ are both flat, yet the neuron is not invariant to x or y: its firing rate is informative about (x, y). The tuning to the two stimulus variables is entangled in a way that the marginal tuning curves cannot reveal.

In the special case that the joint tuning map factorizes into the product of marginal tuning curves, i.e., $h(x, y) = f(x)g(y)$, marginal tuning is sufficient to interpret joint tuning. There is evidence that for certain neurons and certain behavioral variables the joint tuning is close to multiplicative [14]. The generalized linear model (GLM) framework commonly used in neural data analysis [29] makes this assumption when each variable enters through its own additive term: with the exponential link function, adding responses in log firing rate is equivalent to multiplying tuning curves. Interaction terms can be added to a GLM, but their functional form must be specified in advance. When the multiplicative assumption is violated, fitting a GLM with separate terms for each variable leads to misleading conclusions about the selectivity to the stimulus variables (Fig. 1).

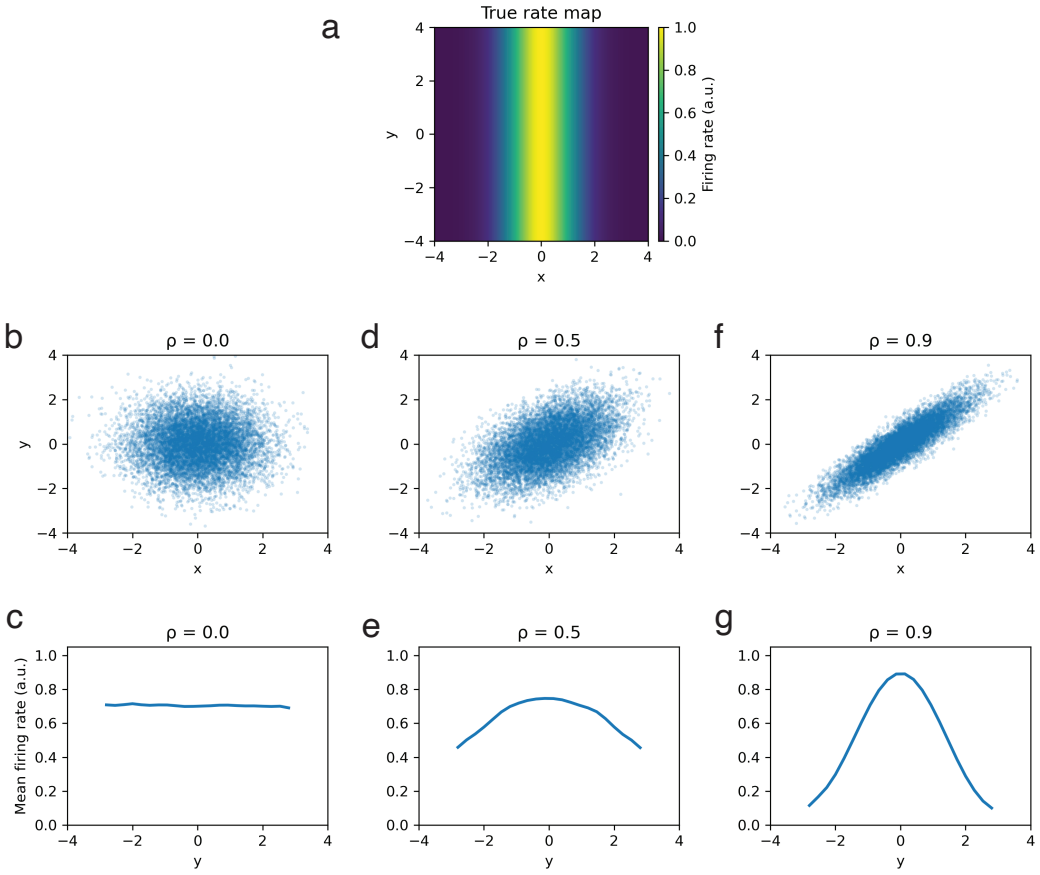


Figure 2: The joint tuning curve of a neuron cannot be adequately described by the marginal tuning curves when stimuli are sampled with correlation. (a) The joint tuning curve of a hypothetical neuron for variables x and y. This neuron is selective to x and invariant to y. (b,c) When the samples of x and y are uncorrelated, the empirically computed marginal tuning for y recovers the ground truth tuning. (d,e) When the samples of x and y are correlated, the empirically computed marginal tuning for y shows selectivity. (f,g) Same as (d,e), with stronger correlation between x and y.

Correlated sampling of the stimuli further complicates the interpretation of marginal tuning curves. Correlated stimulus variables are common. In spatial navigation, one notable example is the animal's head

direction and movement direction [30]; another is time and distance traveled when the animal runs on a linear track. Suppose that a neuron is selective to x and completely invariant to y (Fig. 2). Because of the correlation in stimulus sampling, the marginal tuning curve for y still shows substantial selectivity (see Methods). Relying on the marginal tuning curves alone, we would incorrectly conclude that the neuron is tuned to y, although its response is completely invariant to y. Analyzing the joint tuning directly avoids this error.

2.2 A method for analyzing the invariance and selectivity of neurons

The results above show that, when analyzing the selectivity and invariance of neural responses, it is necessary to examine the joint tuning beyond the marginal tuning curves. A simple way to assess whether the response is invariant to a stimulus variable x is to vary x and examine the tuning to the remaining variables. With two variables, this amounts to examining the tuning curve for y *conditioned on* different values of x. If the tuning curve for y does not change with x, the response of the neuron is invariant to x. Otherwise, the response depends on x.

We quantify the contribution of each variable from the joint tuning map, based on this intuition. To quantify the contribution of x, we ask how much the neuron's tuning over y changes across values of x. We slice the joint tuning map at each value of x to obtain tuning curves over y, and compute the Pearson correlation between tuning curves at different values of x. If these tuning curves are all similar, x contributes little; if they differ, x shapes the response. The contribution of y is computed symmetrically by slicing along the other axis. Because Pearson correlation is invariant to the scale and offset of each tuning curve, the measure captures changes in the shape of the tuning and is insensitive to pure gain or baseline modulation.

Tuning curves estimated from finite data are noisy, and noise lowers the correlation between tuning curves even when the underlying tuning is identical. To measure this effect, we estimate the joint tuning map separately from two halves of the data and correlate each tuning curve in one fold with each tuning curve in the other fold (Fig. 3, top). Correlations between tuning curves at different values of x quantify how much the tuning changes with x. Correlations between tuning curves at the same value of x quantify the reliability of the estimate: one minus this reliability is the value the measure takes for a neuron that is invariant to x (the finite-sampling floor; see Methods). In simulations with known ground truth, the estimator recovers the contribution of each variable, and on an axis to which the neuron is invariant it returns the finite-sampling floor (Fig. 3, bottom; Supplementary Fig. S4).

2.3 Model units in deep network models based on vision exhibit entangled spatial tuning

While many studies suggest that path integration is important for the formation of cognitive maps in the brain [15, 21, 22], recent studies reported place-cell-like responses in deep neural networks trained or

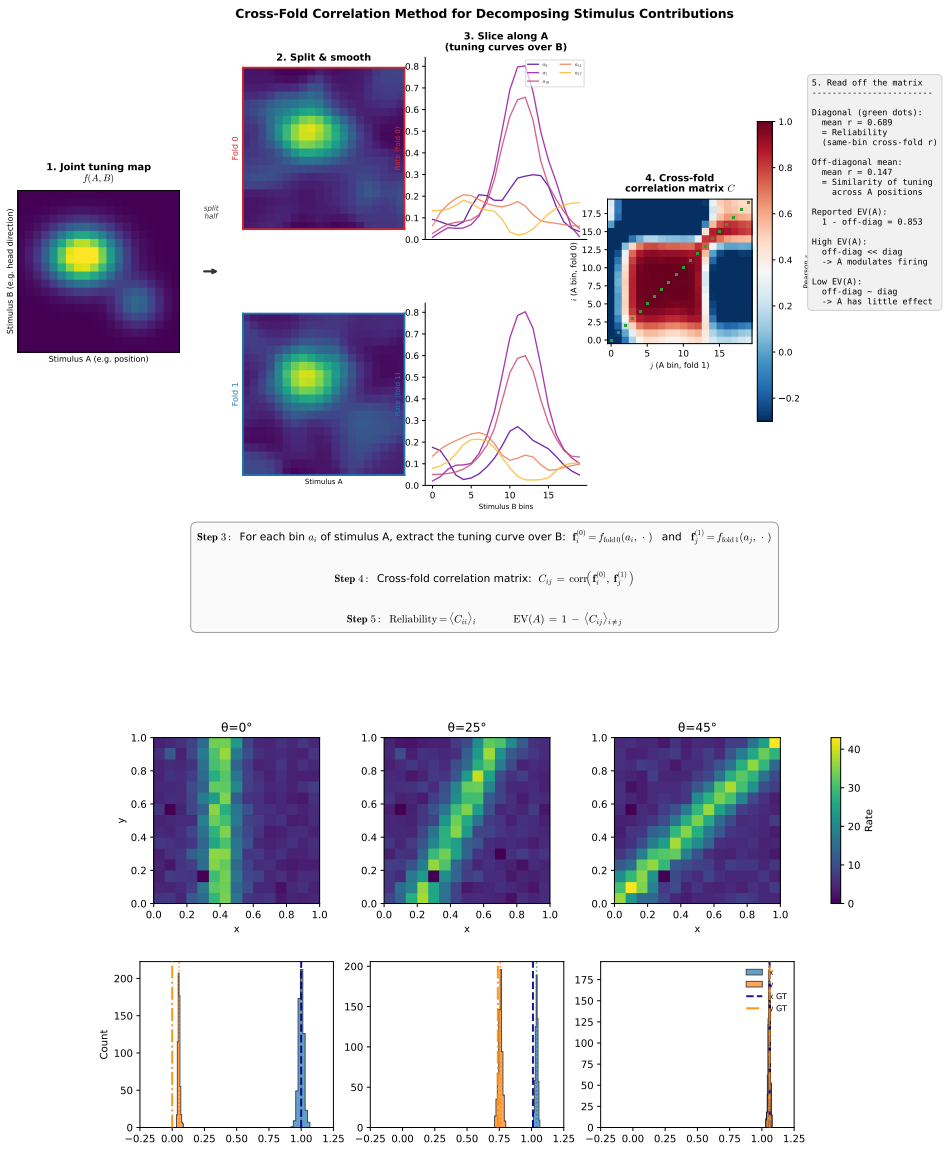


Figure 3: The contribution of each stimulus variable is quantified with the cross-fold correlation method. (Top) Schematic of the estimation procedure for the contribution of stimulus A. The data are split into two halves and a joint tuning map is estimated from each fold. Each map is sliced at every value of A to obtain tuning curves over the other variable B; the tuning curve at a_i in fold 0 is correlated with the tuning curve at a_j in fold 1 to form the cross-fold correlation matrix C_{ij} (Eq. 1). The mean diagonal is the split-half reliability of the maps; the contribution of A is $\text{EV}(A) = 1 - \langle C_{ij} \rangle_{i \neq j}$ (Eq. 2). (Bottom) Validation on simulated Poisson neurons whose log firing rate is a Gaussian ridge rotated by $\theta = 0°$, $25°$ and $45°$, sampled on a 15×15 grid with $1{,}000$ samples (top row: estimated joint tuning maps from one simulation). Histograms show $\text{EV}(x)$ (blue) and $\text{EV}(y)$ (orange) over 600 simulated data sets, each split once into two folds; dashed lines mark the values computed from the noiseless tuning map. At $\theta = 0°$ the neuron depends on x only: $\text{EV}(x) = 1.00$, and $\text{EV}(y) = 0.05$, which equals the finite-sampling floor $1 - \langle C_{ii} \rangle$ of the split-half estimate. At $\theta = 25°$ the estimates are $\text{EV}(x) = 1.04$ and $\text{EV}(y) = 0.76$ (noiseless values 1.01 and 0.74), and at $\theta = 45°$ both are 1.06 (noiseless value 1.06). EV exceeds 1 when tuning curves at distant positions are anticorrelated. The size of the floor depends on how densely the joint map is sampled (Supplementary Fig. S4).

pretrained to perform vision tasks alone [28, 27]. Gornet and Thomson [27] reported that training a

virtual navigation agent to predict the next image frame led to the formation of cognitive maps, with units in certain hidden layers selective to the agent's location, similar to place cells. Luo et al. [28] analyzed the responses of units in ResNet (pretrained on image classification, or untrained) in a spatial environment and reported that a large fraction of units throughout the network exhibit spatial selectivity to location or head direction. These studies [28, 27] examined the marginal tuning to these spatial variables separately, without examining their joint tuning. We examine the joint tuning of units in the types of deep network models studied in [28, 27], and then compare them with neurons in the rodent spatial navigation system.

Model units in networks trained on predictive coding exhibit entangled spatial tuning Inspired by prior work on predictive coding [31, 32], Gornet and Thomson [27] trained a virtual navigation agent, implemented as a deep neural network, to predict the next image frame in a sequence of images as the agent moves in space (Fig. 4a), and found place-cell-like responses in its intermediate layers.

We trained this model in a Minecraft environment similar to that of [27] (Methods) and examined the responses of units in different layers of the trained model. Because the head direction of the agent varies, we obtain the response of each unit for every combination of 2-D location and head direction. Using established place-field criteria [33], a fraction of units in the intermediate layers are classified as putative place cells (Fig. 4i), consistent with [27]. The joint tuning to location and head direction of these units is complex and entangled (Fig. 4c–h). When conditioned on different head directions, the location tuning maps change (Fig. 4e,h): the location selectivity of these units is not invariant to head direction.

Quantifying the contributions of location and head direction with the method above, location and head direction contribute about equally to the responses of most units (Fig. 4j; pre-prediction layer: median $\mathrm{EV}_{xy} = 0.93$, median $\mathrm{EV}_{hd} = 0.92$, $n = 8{,}000$ units, 54% with $\mathrm{EV}_{xy} > \mathrm{EV}_{hd}$). Units classified as putative place cells (red) fall in the same region as the other units. The same holds in the other layers we analyzed, from the first convolutional layer to the output of the attention blocks (Supplementary Fig. S1).

Pretrained convolutional neural networks (CNNs) also exhibit entangled spatial tuning We next investigate deep neural networks pretrained to perform image classification. Convolutional neural networks (CNNs) are a popular modeling framework for how the brain processes visual information [35, 36, 37]. Luo et al. [28] proposed using CNNs to study the navigation system in the brain. They modeled the visual inputs of a virtual agent navigating a 2-D open arena similar to the environments of rodent neurophysiology experiments. These CNNs are pretrained on visual tasks and receive only visual inputs, with no input about the agent's velocity. Luo et al. [28] reported that many units throughout the network exhibit place-cell-like responses as well as head direction tuning.

We analyzed a ResNet-50 [38] pretrained on image classification in the same Minecraft environment used for the predictive coding network, which is similar to the environment of [28]. Consistent with [28], a fraction of units exhibit putative place fields in 2-D space. Luo et al. [28] also reported that many units exhibit strong head direction tuning and that a subset exhibit conjunctive place and HD tuning. In our analysis of marginal tuning, few units are tuned to HD, and consequently few units are tuned to both HD and location. In the code released with [28], the criterion for head direction cells does not properly capture selectivity to head direction: applying that criterion to our network also classifies many units

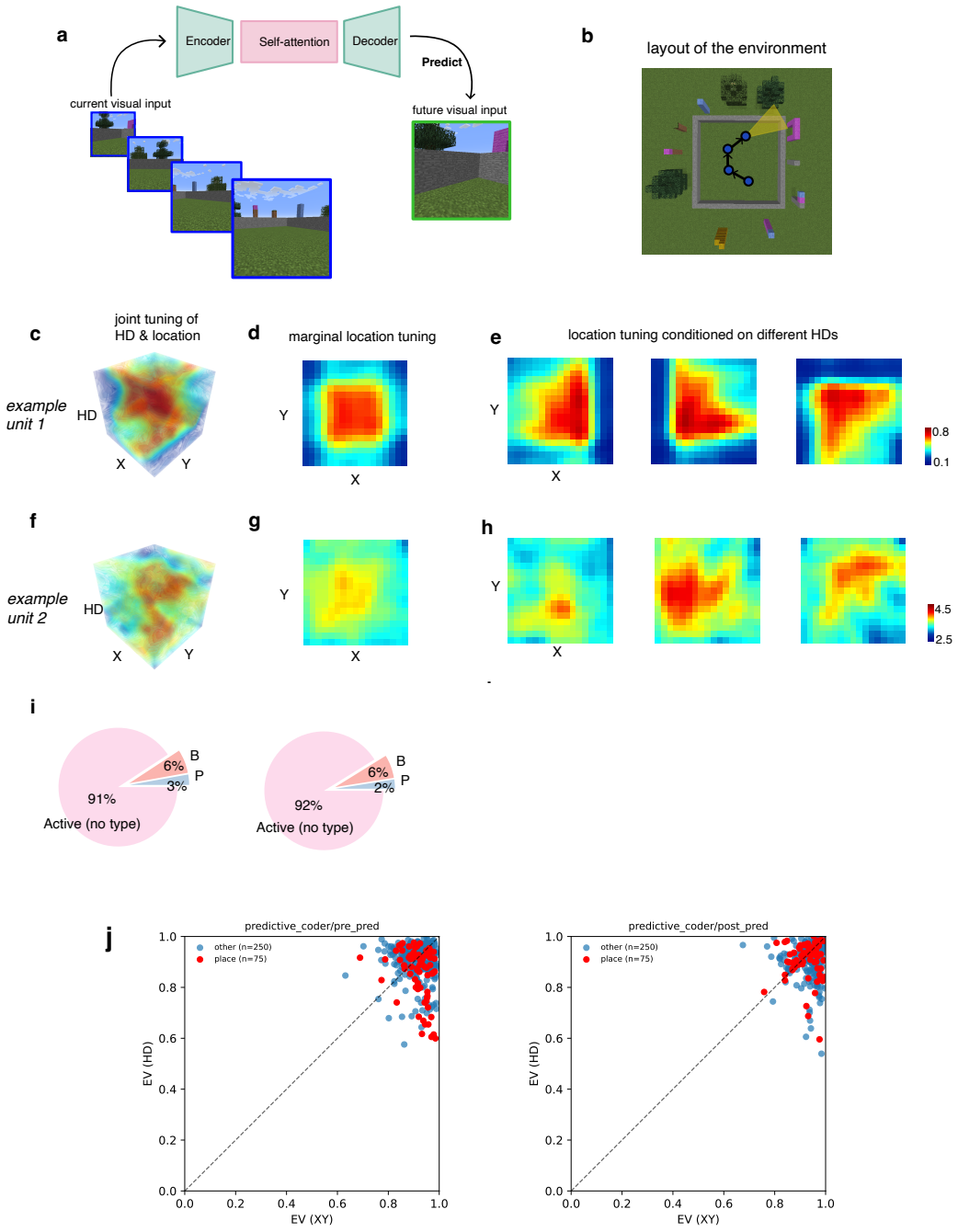
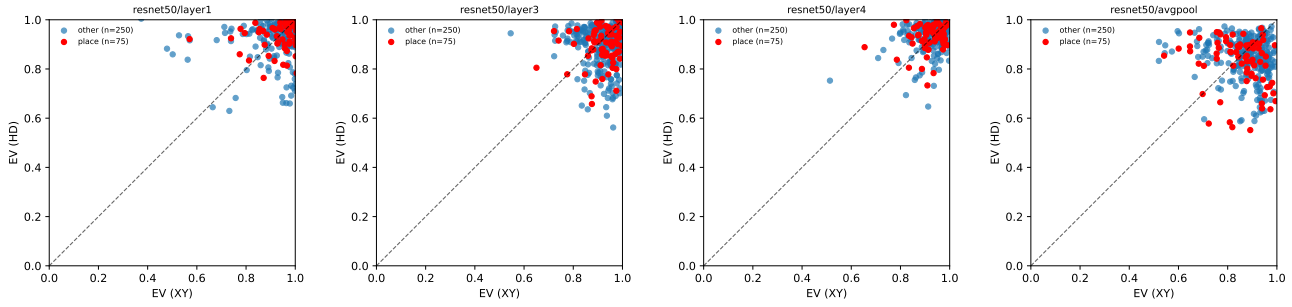


Figure 4: Model neurons in deep networks trained to predict the next image frame exhibit entangled tuning to location and head direction. (a) Schematic of the modeling framework of [27], adapted from [34]. (b) Layout of the virtual environment. (c–h) Two example model units classified as place cells: joint tuning to location and HD (c,f), marginal location tuning (d,g), and location tuning conditioned on different head directions (e,h). The marginal 2-D map shows a place field, but the location tuning changes with HD. (i) Fraction of model units classified as place (P) or border (B) cells in two layers of the model. (j) Contribution of location (EV_{xy}) versus head direction (EV_{hd}) (Eq. 2) for units in the pre-prediction (left) and post-prediction (right) latent layers; red points are units classified as place cells; the dashed line marks $\mathrm{EV}_{xy} = \mathrm{EV}_{hd}$.

as head direction cells. The difference in the prevalence of head-direction-tuned units between the two studies therefore arises from the classification criterion rather than from the virtual environment.



(a) `layer1` (b) `layer3` (c) `layer4` (d) `avgpool`

Figure 5: Model neurons in CNNs trained on image recognition exhibit entangled tuning to location and head direction. (a–d) Contribution of location (EV_{xy}) versus head direction (EV_{hd}) (Eq. 2) for individual units in four layers of an ImageNet-pretrained ResNet-50 (`layer1`, `layer3`, `layer4`, and `avgpool`). Red points are units classified as putative place cells; the dashed line marks $\mathrm{EV}_{xy} = \mathrm{EV}_{hd}$. In every layer the population lies on the diagonal, with median EV_{xy} and EV_{hd} both ≈ 0.9: location and head direction contribute equally to the responses, unlike place cells and head direction cells in the brain. Results for a VGG-16 layer and an untrained ResNet-50 are shown in Supplementary Fig. S2.

The joint location$\times$HD maps of units classified as putative place cells show firing fields scattered through the 3-D space (Supplementary Fig. S3): a unit is active only for particular combinations of location and HD, as in the network trained on next-frame prediction. Across all layers, location and head direction contribute about equally to the responses of individual units (Fig. 5 for four layers of ResNet-50; Supplementary Fig. S2 for VGG-16 [39] and an untrained ResNet-50).

Together, these results reproduce key findings of recent studies of spatial tuning in deep neural networks trained on purely visual tasks: in the marginal tuning to 2-D location, a fraction of units have putative place fields. The joint tuning, however, shows that location and head direction are strongly entangled in these models. For most units, location and head direction contribute about equally to the response, and the models contain no distinct populations that encode head direction and location separately.

2.4 Distinct selectivity and invariance of place cells and head direction cells in rodent brain

We next analyze neurons recorded in the rodent spatial navigation system and compare them to the deep network models. We analyze two datasets: CA1 place cells recorded while rats navigated a 2-D open arena [40], and head direction cells recorded from the postsubiculum of mice [41]. Beyond the selectivity that defines each cell type, we ask whether these cells are invariant to the other spatial variables, e.g., whether place cells are invariant to head direction.

Place cells in the rat hippocampus are largely invariant to head direction We computed the 2-D firing rate maps of putative excitatory CA1 neurons from [40] ($n = 551$ neurons from four rats). Consistent with previous reports, many of these neurons are selective to location in the open arena, with compact place fields, and most do not exhibit strong marginal tuning to head direction. As shown above (Fig. 1), a flat marginal tuning curve does not establish that a response is invariant to a variable. We therefore

computed the joint tuning map for location and HD, a 3-D map for each neuron (Fig. 6a,b). When conditioned on head direction, the location of the place field is stable. The peak firing rate of the place field varies modestly across head directions; because the correlation measure is insensitive to gain, this modulation does not contribute to EV_{hd}.

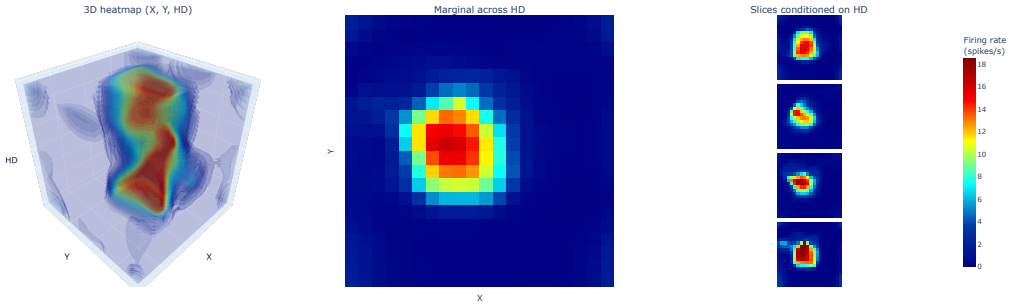
We quantified the contributions of location and head direction with the method above (Fig. 3). Location contributes far more than head direction (Fig. 6c): EV_{xy} exceeds EV_{hd} for 99% of neurons (median $\mathrm{EV}_{xy} = 0.97$, median $\mathrm{EV}_{hd} = 0.49$). The finite-sampling floor of EV_{hd}, $1 - \langle C_{ii} \rangle$, has a median of 0.19 across neurons, so the remaining EV_{hd} reflects changes in the shape of the location tuning across head directions. Normalized by the reliability of the maps, $(\langle C_{ii} \rangle - \langle C_{ij} \rangle_{i \neq j}) / \langle C_{ii} \rangle$, the contribution of head direction has a median of 0.37, compared with 0.93 for location.

The simultaneously recorded populations confirm this result. Location is decoded accurately (mean absolute error 14.3 cm, averaged over the four rats, in a 200 cm arena; 57.1 cm for the chance decoder), whereas head direction is decoded poorly (mean absolute error $69.5°$; $81.9°$ for the chance decoder and $90°$ for uniform random guessing; Fig. 6d,e).

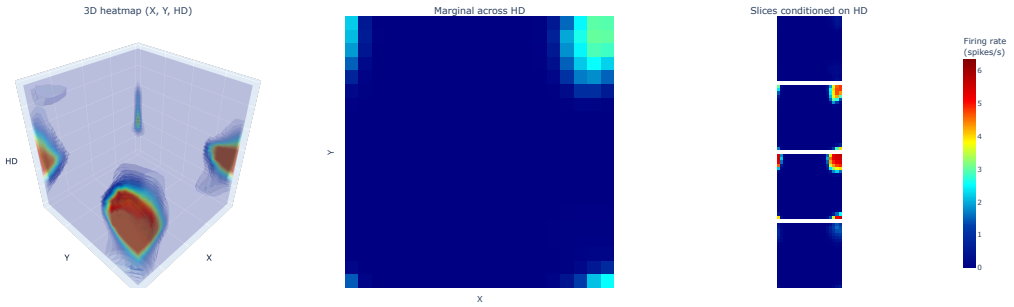
Rodent head direction cells are largely invariant to the animal's location We next analyzed head direction (HD) cells recorded from the postsubiculum of mice [41]. In the 3-D joint tuning maps for head direction and location (Fig. 7a,b), the head direction tuning curves are similar when the animal is in different quadrants of the environment. Head direction contributes far more to the responses than location (Fig. 7c): across $n = 1{,}489$ HD cells from 30 sessions, EV_{hd} exceeds EV_{xy} for 94% of cells (median $\mathrm{EV}_{hd} = 0.92$, median $\mathrm{EV}_{xy} = 0.67$). The finite-sampling floor of EV_{xy} has a median of 0.32, reflecting the sparse sampling of the 400×90 location$\times$HD bins within a session; normalized by reliability, the contributions are 0.85 for head direction and 0.52 for location. Decoding from the simultaneously recorded populations ($n = 31$ sessions) gives a small error for head direction (mean absolute error $12.3°$; $85.2°$ for the chance decoder). Location cannot be decoded: the decoding error (32.9 cm in an 80 cm arena) exceeds that of the chance decoder that always reports the most occupied location (23.3 cm).

Place cells and head direction cells in rodents thus show a double dissociation. The response of a place cell depends strongly on the location of the animal and is largely invariant to its head direction, whereas the response of a head direction cell depends strongly on head direction and only weakly on location.

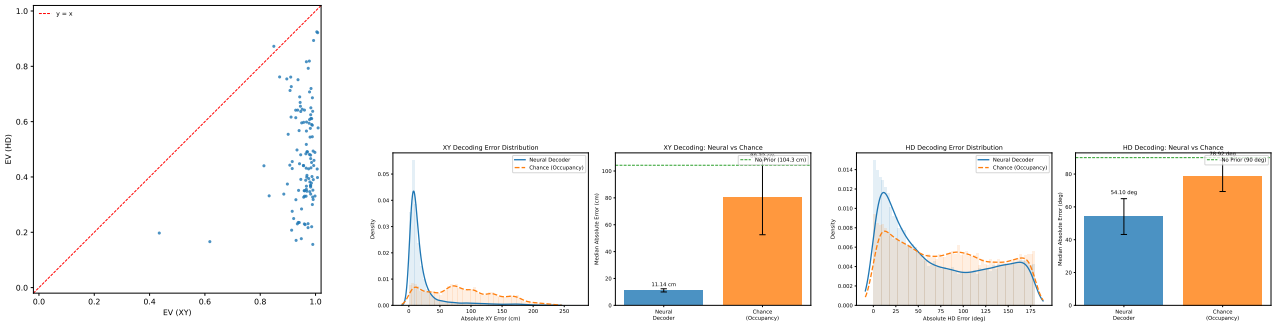
Comparison of the deep network models and the rodent hippocampus The joint tuning to head direction and location reveals a large discrepancy between model units and spatial cells in the brain. Model units exhibit entangled selectivity to head direction and location, whereas place cells and head direction cells show disentangled selectivity. These models based on visual inputs alone therefore do not account for the key properties of the spatial cells found in the brain, and the joint tuning of multiple variables provides a more informative signature than marginal tuning for testing such models.



(a)



(b)

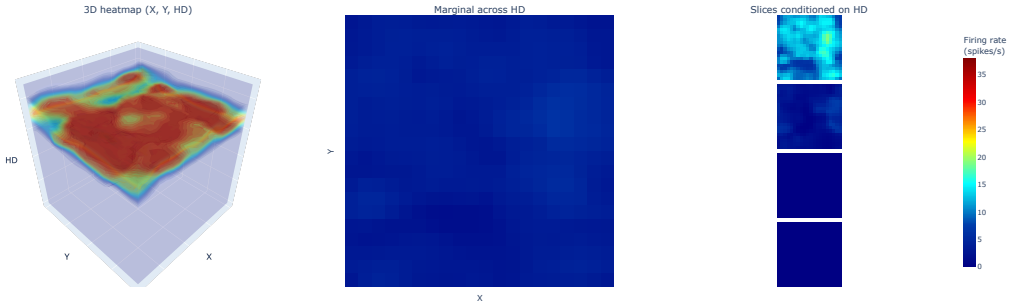


(c) (d) (e)

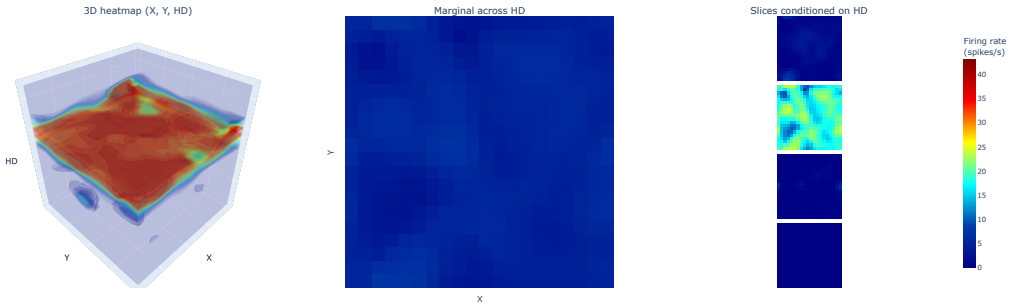
Figure 6: Place cells in the rat hippocampus are selective to location and largely invariant to head direction. (a) An example place cell (rat 4). Left: the joint tuning map over 2-D location and head direction rendered as a 3-D volume. Middle: the 2-D place field obtained by averaging across HD. Right: location tuning conditioned on four head-direction quadrants. The place field occupies the same location at every head direction. (b) Same as (a) for a second place cell (rat 2). (c) Contribution of location (EV_{xy}) versus head direction (EV_{hd}) for individual neurons (30 randomly chosen neurons per rat shown; $n = 551$ neurons from 4 rats in total). The dashed line marks $\mathrm{EV}_{xy} = \mathrm{EV}_{hd}$; 99% of neurons lie below it. (d) Decoding of 2-D location from the simultaneously recorded population (blue) compared with a chance decoder that always reports the most occupied bin (orange). Bars show the median absolute error over all decoded time bins; error bars, s.d. across rats. (e) Same for head direction. The dashed line marks the expected error of uniform random guessing.

2.5 Head direction cells are only weakly tuned to movement direction, and thus are inadequate for providing a path integration signal

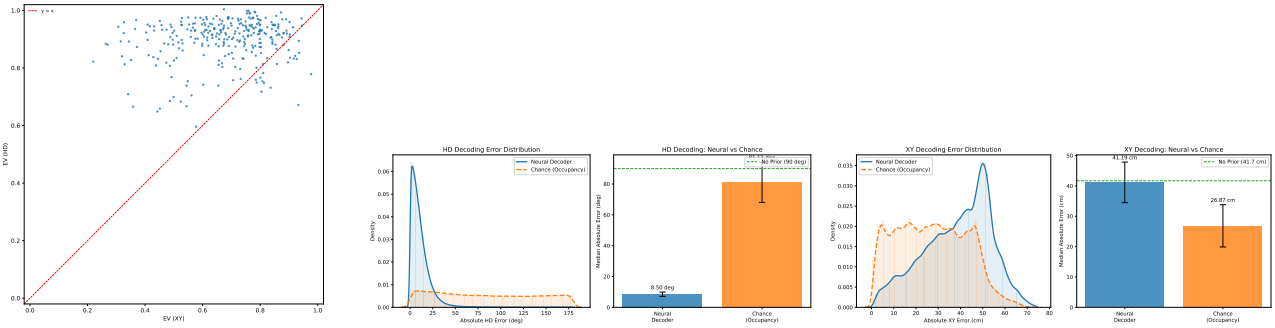
The analysis of joint tuning applies beyond the comparison of deep network models and the hippocampus. We next use it to study a question that is central to understanding grid cells [3, 6]. A popular



(a)



(b)



(c) (d) (e)

Figure 7: Head direction cells in the mouse postsubiculum are selective to head direction and largely invariant to location. (a,b) Two example HD cells, shown as in Fig. 6a,b: the joint location×HD tuning volume (left), the location map averaged across HD (middle), and location maps conditioned on four HD quadrants (right). Activity fills the whole arena at the preferred head direction and is absent elsewhere, i.e., the HD tuning does not depend on location. (c) Contribution of location versus head direction for individual HD cells (10 randomly chosen cells per session shown; $n = 1{,}489$ cells from 30 sessions in total); 94% of cells lie above the diagonal. (d) Decoding of head direction from the simultaneously recorded HD-cell population compared with a chance decoder that always reports the most occupied bin. Bars show the median absolute error over all decoded time bins; error bars, s.d. across sessions. (e) Same for 2-D location.

class of computational models proposes that the hexagonal grid firing pattern results from path integration [21, 15, 42, 22]. This class of models requires that grid cells receive input about the animal's velocity (i.e., movement direction and speed). Neurons that encode movement speed have been identified in the entorhinal cortex and other areas [10, 43], but neurons that selectively encode movement direction have

not been identified, and head direction is often assumed to stand in for movement direction. Although the two are correlated, a previous study [30] found that (i) head direction is more strongly encoded in head direction cells than movement direction, and (ii) head direction and movement direction are sufficiently distinct that head direction would be an insufficient input for generating grid patterns in path integration models. It has remained unclear whether head direction cells are tuned to movement direction (MD) once the correlation between head and movement directions is taken into account, and whether the head direction population encodes a movement direction signal accurate enough for path integration.

We re-analyzed the head direction cells of [41] to address these questions. A substantial fraction of these cells show marginal tuning to MD, which could reflect either selectivity to MD or the correlation between MD and HD. To separate the two, we examined the 2-D joint tuning to HD and MD. In example neurons, the MD tuning conditioned on HD is weak (Fig. 8a,b). The joint map is well populated away from the $\mathrm{MD} = \mathrm{HD}$ diagonal: in these open-field sessions head direction and movement direction are only moderately coupled (Fig. 8c; pooled over sessions, MD lies within 30° of HD for 25% of the time above the 1 cm/s speed threshold, compared with 17% expected for independent variables), so the conditioned tuning curves are estimated from many distinct HD–MD combinations. Quantified with our method, MD contributes little to the responses: EV_{hd} exceeds EV_{md} for every one of $n = 1{,}489$ HD cells from 30 sessions (median $\mathrm{EV}_{hd} = 0.95$, median $\mathrm{EV}_{md} = 0.18$; Fig. 8d). The finite-sampling floor of EV_{md} is small (median 0.04), and normalized by reliability the contribution of MD has a median of 0.14.

Decoding from the population confirms that the MD signal in head direction cells is weak (Fig. 8e): the mean absolute error of the decoded MD across $n = 31$ sessions is 71.7°, compared with 89.8° for the chance decoder and 90° for uniform random guessing. The MD signal in head direction cells is therefore too weak to support accurate path integration, and the neural signal that encodes the movement direction needed for path integration has yet to be identified.

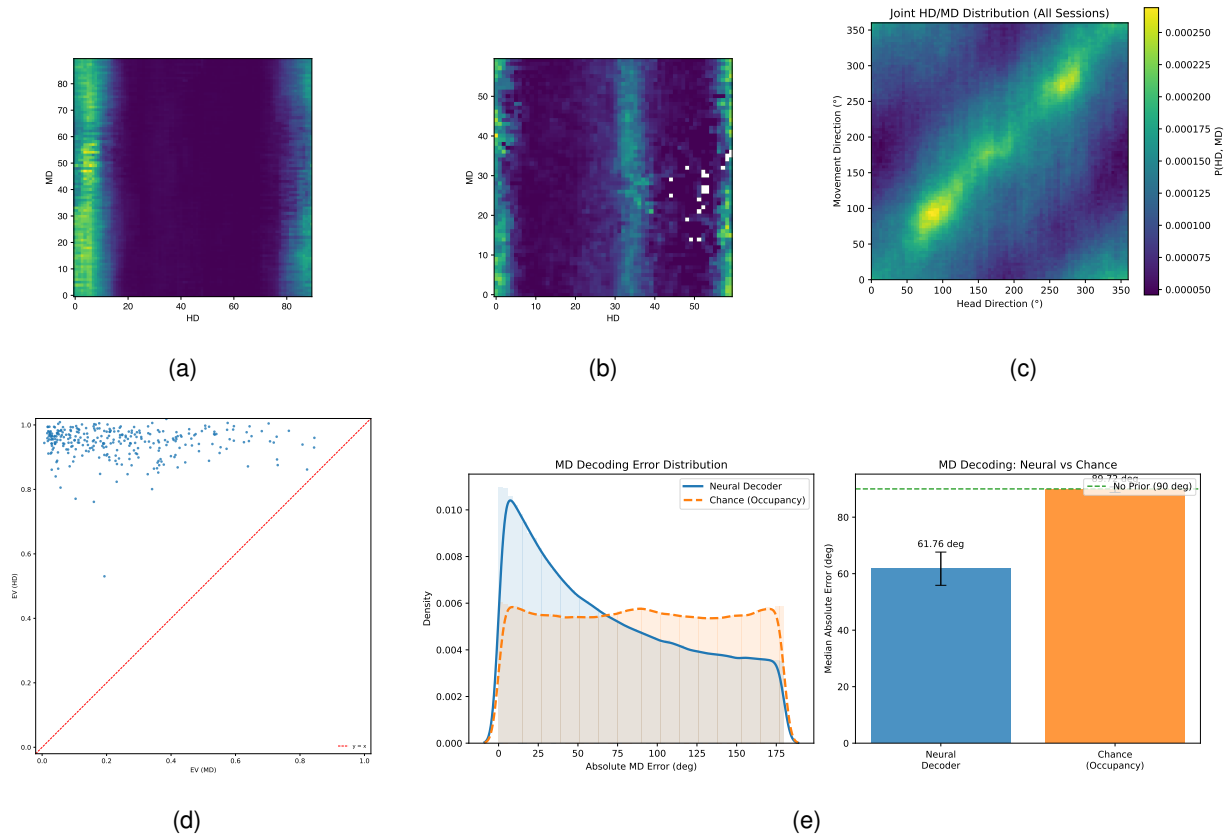


Figure 8: The MD signal encoded in the head direction population is weak. (a,b) Joint HD×MD tuning maps (4° bins) of two example HD cells. Firing is confined to a vertical band, i.e., it depends on HD but not on MD. (c) Joint distribution of HD and MD pooled over all sessions (time above the 1 cm/s speed threshold). The two variables are only moderately coupled, so the joint tuning maps in (a,b) are well sampled off the diagonal. (d) Contribution of MD versus HD for individual HD cells (10 randomly chosen cells per session shown; $n = 1{,}489$ cells from 30 sessions in total); every cell lies above the diagonal. (e) Decoding of MD from the HD-cell population compared with a chance decoder that always reports the most occupied bin.

3 Discussion

Joint tuning to multiple stimulus variables contains information that the standard analysis of marginal tuning curves misses, and the selectivity and invariance of a neural code need to be studied together. The joint tuning of neurons is often studied with GLMs [29, 14], which allow the response variance to be partitioned among many variables. A GLM with separate terms for each variable assumes that the joint tuning factorizes into the product of the marginal tuning curves, an assumption that may be violated in practice. When many behavioral variables must be considered simultaneously, a GLM is an effective approach, since more flexible models cannot be fit with the available data. When only a handful of variables are relevant, our approach provides an alternative that makes no assumption about how the variables interact and can reveal structure in the code that a GLM with separate terms cannot. Our measure targets changes in the shape of the tuning across values of a variable; gain modulation is invisible to it and must be assessed separately.

The joint tuning of spatial cells in rodents reveals their selectivity and invariance. Place cells are largely

invariant to the head direction of the animal: the location of the place field is stable across head directions, while the shape of the field changes modestly and its peak rate varies. Head direction cells are largely invariant to location. The invariance of place cells to head direction, and of head direction cells to location, suggests that the code for space is "abstract" and does not derive from selectivity to sensory features alone. Computationally, being selective to one variable while invariant to others supports generalization: the invariance of place cells to head direction allows downstream circuits to extract spatial information with the same readout, without adjusting the weights according to the current head direction.

Applying our approach to deep network models based on vision tasks [28, 27], the tuning properties of individual units differ drastically from those of place cells and head direction cells in the rodent brain. Individual model units encode entangled representations of head direction and location. Most model units do not exhibit clear marginal tuning to head direction, yet their response is strongly modulated by head direction through its interaction with location. These results hold for both types of deep network models [28, 27]. One interpretation is that these units are selective to visual features, and spatial selectivity emerges because the visual features in the environment depend on location and head direction jointly. Even a unit in an early layer of the network that behaves as an edge detector can exhibit location and HD tuning. Such selectivity is substantially different from that of head direction cells and place cells: the network units are much less invariant to head direction and location than place cells and head direction cells in rodents. These results reveal a major gap in current deep neural network models trained on vision tasks as accounts of the tuning properties of the rodent spatial navigation system. An open question is what additional ingredients would make model units selective to location and invariant to head direction, as in rodents.

Our approach applies to the joint tuning of individual neurons to a small number of behavioral variables (2 or 3) without assumptions about how the tuning to different variables interacts (such as the multiplicative interaction of a GLM with separate terms for each variable). Applied to head direction cells, it shows that they are largely invariant to movement direction, so head direction cells do not supply the accurate movement direction signal required for path integration [15]. The same technique can address the selectivity of place cells on linear tracks to location, time, and distance traveled. Hippocampal neurons on linear tracks have been proposed to encode location, time, distance traveled, or a combination of these [44, 45], and these factors are hard to disentangle because they are highly correlated in experiments. Our approach estimates the contribution of each factor from the joint tuning.

Our results also bear on the study of mixed selectivity [12], which has been reported in many brain areas and tasks. Methodologically, our approach quantifies mixed selectivity through the joint tuning and the contribution of each stimulus variable. Our analysis of the deep network models further shows that mixed selectivity can arise from analyzing the wrong set of stimulus variables. In the deep network models examined here, many units in the first layer are oriented filters that encode the orientation of edges. Analyzed in terms of spatial variables, these units exhibit complex nonlinear mixed selectivity, because the mapping between the visual feature and the spatial variables (location and head direction) is highly nonlinear: an edge of a given orientation appears in the receptive field of the unit only for particular combinations of location and head direction. Mixed selectivity in neural systems has been

proposed to carry functional advantages for coding [12, 13]; the example above shows that it can also arise from misidentifying the encoded features, which calls for care in interpreting responses that appear mixed selective.

This paper focuses on the response properties of individual neurons. An important question is what the observed selectivity and invariance imply at the population level. The tuning properties of individual neurons constrain population-level properties because they define the projection of the neural manifold onto the individual neuronal axes [7], but this relationship has mainly been studied for a single stimulus variable. Understanding the relationship between the joint tuning to multiple variables and the structure of the neural manifold is an important direction for future research.

4 Methods

4.1 Analysis of marginal tuning curves under correlated stimulus sampling

Here we show how correlated sampling of the stimuli produces spurious tuning in the standard analysis of tuning curves. Consider the tuning of a neuron to variables x and y, with joint tuning $h(x, y)$.

Assume that the marginal sampling distribution over x is uniform and that the joint tuning factorizes, $h(x, y) = f(x)g(y)$. The marginal tuning curve for x estimated by the standard averaging procedure is

$$\hat{f}(x) = \int p(y|x)h(x, y)dy = \int p(y|x)f(x)g(y)dy = f(x) \int p(y|x)g(y)dy.$$

If x and y are independent, the procedure recovers $f(x) \int p(y)g(y)dy$, which is proportional to $f(x)$. Otherwise, through the $p(y|x)$ term, the estimated marginal tuning curve for x depends on $g(y)$.

Consider the special case where the neuron is invariant to x, i.e., $h(x, y) = h(y)$ is a narrow Gaussian function, so that the true tuning curve for x is constant. If only $y = x$ is sampled (perfectly correlated sampling of the two stimuli), the estimated tuning curve is

$$\hat{f}(x) = \int p(y|x)h(y)dy = \int \delta(y - x)h(y)dy = h(x).$$

Relying on the marginal tuning curves, we would incorrectly conclude that the neuron is strongly tuned to x.

Simulation parameters for the marginal confound and GLM demonstrations (Figs. 2 and 1)

For the marginal confound demonstration (Fig. 2), we simulated a neuron with tuning $h(x, y) = \exp(-x^2/2\sigma_x^2)$ with $\sigma_x = 1.0$ (arbitrary units), selective to x and invariant to y. We drew $N = 300$ samples with x uniform on $[-4, 4]$ and $y \sim \mathcal{N}(x, \sigma_{\text{cond}}^2)$ clipped to $[-4, 4]$. Three correlation levels were obtained with $\sigma_{\text{cond}} \in \{4.8, 2.1, 0.6\}$, corresponding to weak ($\rho \approx 0$), moderate ($\rho \approx 0.6$), and strong ($\rho \approx 0.9$) correlation. Marginal tuning curves for y were computed in 30 bins and smoothed with a Gaussian kernel ($\sigma = 1$ bin); bins with fewer than 5 samples were excluded.

For the GLM demonstration (Fig. 1), we simulated Poisson neurons with firing rate $\lambda(x, y) = \exp(\alpha - u^2/2\sigma_u^2)$, where $u = x\cos\theta - y\sin\theta$, $\alpha = \log 5$, $\sigma_u = 1.2$, and $\theta \in \{45°, 60°\}$. Stimuli x and y were drawn independently from $\mathcal{N}(0, 1)$ ($n = 20{,}000$ samples), and spike counts were drawn from a Poisson distribution with rate $\lambda(x, y)$. A Poisson GLM $\log\lambda = \beta_0 + \beta_1 x + \beta_2 y + \beta_3 x^2 + \beta_4 y^2$ was fit by maximum likelihood. The GLM tuning curves $f(x)$ and $g(y)$ were obtained by evaluating the fitted model along each axis with the other variable at its mean.

4.2 Estimating the contribution of individual stimulus variables

For every bin of the joint firing rate map of (x, y), the samples that fall in the bin are randomly split into two equal halves, and each half is averaged to give the two fold maps $f^{(0)}(x, y)$ and $f^{(1)}(x, y)$. With x and y discretized into m and n bins, we write $f^{(0)}_{i,j} = f^{(0)}(x_i, y_j)$ and $f^{(1)}_{i,j} = f^{(1)}(x_i, y_j)$.

To quantify the contribution of x, we ask how much the tuning over y changes across values of x. For each bin x_i, we take the tuning curve over y at $x = x_i$, $\mathbf{f}^{(0)}_i = f^{(0)}(x_i, \cdot)$ and $\mathbf{f}^{(1)}_j = f^{(1)}(x_j, \cdot)$, each a vector of length n. Because these slices are conditioned on x, the measure detects cases where x modulates the response through its interaction with y even when the marginal tuning to x is flat.

We compute the cross-fold Pearson correlation between every pair of these tuning curves:

$$C_{ij} = \mathrm{corr}\left(\mathbf{f}^{(0)}_i,\ \mathbf{f}^{(1)}_j\right). \tag{1}$$

The diagonal entries C_{ii} compare tuning curves at the same value of x across the two folds and quantify the reliability of the tuning estimate. The off-diagonal entries C_{ij} $(i \neq j)$ compare tuning curves at different values of x; if the tuning over y does not depend on x, the off-diagonal correlations are as high as the diagonal ones.

The contribution of x is

$$\mathrm{EV}(x) = 1 - \langle C_{ij} \rangle_{i \neq j}, \tag{2}$$

where $\langle \cdot \rangle_{i \neq j}$ denotes the mean over off-diagonal entries. When x strongly modulates the tuning over y, tuning curves at different values of x differ, the off-diagonal correlations are low, and $\mathrm{EV}(x)$ is high. When x has no influence, the off-diagonal mean equals the diagonal mean and $\mathrm{EV}(x) = 1 - \langle C_{ii} \rangle_i$. This value is the finite-sampling floor of the estimate: it is 0 for noiseless maps and grows as the fold maps become less reliable. We also report the reliability-normalized contribution $\mathrm{EV}_{\mathrm{norm}}(x) = (\langle C_{ii} \rangle_i - \langle C_{ij} \rangle_{i \neq j}) / \langle C_{ii} \rangle_i$, which is 0 for an invariant neuron. Because $\langle C_{ij} \rangle_{i \neq j}$ is negative when tuning curves at distant positions are anticorrelated, $\mathrm{EV}(x)$ can exceed 1. Pearson correlation is invariant to the scale and offset of each tuning curve, so the measure is insensitive to gain and baseline changes across x.

$\mathrm{EV}(y)$ is computed by slicing along the other axis, fixing y and comparing tuning curves over x. Before computing the correlations, each fold map is smoothed with a NaN-aware Gaussian kernel (circular along angular axes) and resampled so that the two axes have comparable numbers of bins; the dataset-specific parameters are given below. Bins in the lowest decile of occupancy are excluded from the correlations. All EV values are computed from a single split, with a fixed random seed.

Simulations for method validation (Figs. 3 and S4)

We simulated Poisson neurons with firing rate $\lambda(s_1, s_2) = \exp\big(\beta_0 + A \exp(-u^2/2\sigma^2)\big)$, with $u = (s_1 - \mu)\cos\theta - (s_2 - \mu)\sin\theta$, $\mu = 0.4$, $\sigma = 0.1$, $\beta_0 = \log 5$ and $A = \log 35 - \log 5$, so that the rate ranges from 5 to 35 spikes per unit time. The orientation θ sets the relative contributions of $x = s_1$ and $y = s_2$: at $\theta = 0^\circ$ the neuron depends on x only, and at $\theta = 45^\circ$ on x and y equally. Stimuli were drawn uniformly on

$[0,1]^2$ and placed at the centers of a 15×15 grid of bins. For each $\theta \in \{0°, 25°, 45°\}$ we generated 600 data sets of $N = 1{,}000$ samples, each with a different random seed, split each data set once into two folds, and computed $\mathrm{EV}(x)$ and $\mathrm{EV}(y)$ without smoothing. The noiseless values were computed with Eq. 2 from the correlation matrix of the noiseless rate map, averaged within each bin. To characterize the finite-sampling floor, we repeated the $\theta = 0°$ simulation (true $\mathrm{EV}(y) = 0$) while varying the number of bins per axis (5–40 at $N = 1{,}000$) and the number of samples ($N = 250$–$8{,}000$ at 15×15 bins), with 100 data sets per condition, recording $\mathrm{EV}(y)$, the floor $1 - \langle C_{ii} \rangle_i$, and $\mathrm{EV}_{\mathrm{norm}}(y)$.

4.3 Neural network models

We evaluated two types of deep networks trained on vision tasks: convolutional neural networks (CNNs) pretrained on image classification, following [28], and a predictive coding network trained to predict the next image frame from the preceding image sequence during navigation [27]. Both were analyzed in the same Minecraft environment.

Environment and images The environment is a 17×17 block area with distinctive features along the borders to reduce visual aliasing, similar in size to the environment of [28]. First-person images were collected with the Malmo platform [46] at each of the 289 block positions. At each position we rendered a panorama of 90 head directions ($4°$ steps) at a camera pitch of $0°$, and a second set of panoramas with the same 90 head directions at each of 4 camera pitch levels.

CNNs We analyzed an ImageNet-pretrained ResNet-50 [38] (`layer1`, `layer3`, `layer4` and `avgpool`), the `avgpool` layer of a randomly initialized ResNet-50, and layer `features.9` of an ImageNet-pretrained VGG-16 [39], with the torchvision weights. None of the networks were trained further on the environment. Images were resized and center-cropped to 224×224 pixels and normalized with the ImageNet channel means and standard deviations.

Predictive coding network We implemented the model of Gornet and Thomson [27]: a UNet-style encoder–decoder in which the encoder and decoder are ResNets with skip connections, with two self-attention blocks (each followed by a feed-forward block) in the latent space between them. The model receives a sequence of 10 first-person images from a trajectory of an agent moving through the Minecraft environment and is trained to predict the next image in the sequence. To generate training trajectories, the agent started at a random position and updated its movement direction by randomly sampling small changes to its angular velocity, executing minimal turns as needed to avoid obstacles. The model was trained for 200 epochs by gradient descent with Nesterov momentum [47], a weight decay of $5 \cdot 10^{-6}$, and a learning rate of 10^{-1} with OneCycle scheduling [48]. We analyzed four layers: the first convolutional layer (`conv1`) and first downsampling block (`down1`) of the encoder, the encoder output before the attention blocks (pre-prediction latent), and the output of the second attention and feed-forward block (post-prediction latent). To obtain the response to a single view, each image was repeated as a sequence of 10 identical frames and the latent at the last time step was taken.

Unit responses and EV For layers with more than 8,000 units, a fixed random subset of 8,000 units was analyzed. Responses to the 0°-pitch panoramas define each unit's location$\times$HD map (289×90) used for cell classification. Responses to the pitch panoramas were used for EV: the two folds are the mean responses over pitch levels $\{1, 3\}$ and $\{2, 4\}$, so that the two fold maps are computed from different images of each (location, head direction) bin. The fold maps were smoothed with a Gaussian kernel (s.d. 1 bin, circular along head direction), and the head-direction axis was upsampled by a factor of 5 before computing EV_{xy} and EV_{hd}.

Place cell classification We classified units with place-field criteria adapted from Tanni et al. [33], with stability assessed across head directions in place of recording time. The location$\times$HD map of each unit was rescaled to $[0, 1]$ and summed over head directions, and the resulting location map was thresholded at its mean. Candidate fields are contiguous regions (four-neighbor connectivity) of at least 15 bins whose peak is at least half the maximum of the map. A candidate is retained as a place field if it covers no more than half of the map and if, within the field, the location maps computed from the first and second halves of the head directions correlate with $r \geq 0.3$ (Pearson correlation over field bins above half the threshold, at least 9 bins). A unit with at least one place field is a putative place cell; units that also pass the border-cell (border score > 0.5) or head-direction-cell (directional score > 0.47) criteria are excluded. The pyramidal waveform criterion of [33] has no analogue for model units and is not used.

4.4 Neural data

Place cells We analyzed the recordings of Pfeiffer and Foster [40]. Four adult male Long-Evans rats performed an open-field spatial memory task in a 2×2 m black arena enclosed by 30 cm walls, with 36 identical, evenly spaced reward wells flush with the floor. On each trial, after consuming a reward at a randomly selected well, the rat navigated back to a fixed home well for a second reward; the home well changed each day. Each session contained about 30 trials, giving broad coverage of the arena. Wells were filled silently via tubing beneath the floor, and no explicit cue signaled which well was filled. Each rat was implanted with a microdrive array targeting dorsal CA1. Units were identified by manual clustering of spike waveform amplitudes; only well-isolated units were included, and putative interneurons were excluded by spike width and mean rate. Position and head direction were tracked at 60 Hz with two head-mounted LEDs recorded by an overhead camera. We analyzed all putative excitatory units of the first open-field session of each rat ($n = 551$).

Spikes were binned into 250 ms time bins. Location was binned into 20×20 bins (10 cm) and head direction into 60 bins (6°). Time bins in which the animal moved slower than 5 cm/s were excluded, and location$\times$HD bins with fewer than 5 time bins were discarded. The fold maps were smoothed with a Gaussian kernel (s.d. 1 bin along location and 2 bins along head direction), and the head-direction axis was resampled to match the 400 location bins.

For decoding, we used a memoryless Bayesian decoder with independent Poisson neurons and a uniform prior. For each state bin s (a head-direction bin or a 2-D location bin), the tuning $f_i(s)$ of each

neuron was estimated from occupancy-normalized firing rates. Given spike counts $\mathbf{n}_t = (n_{t,1}, \ldots, n_{t,N})$ in a time window of duration τ, the posterior is

$$P(s \mid \mathbf{n}_t) = \frac{\mathcal{L}(s)}{\sum_{s'=1}^{M} \mathcal{L}(s')}, \tag{3}$$

where M is the number of state bins and the Poisson likelihood is

$$\mathcal{L}(s) = \left(\prod_{i=1}^{N} [f_i(s)\,\tau]^{n_{t,i}} \right) \exp\left(-\tau \sum_{i=1}^{N} f_i(s) \right). \tag{4}$$

The decoded state is the posterior maximum. For the place cells we used $\tau = 250$ ms windows, excluded time bins with speed below 5 cm/s, binned head direction into 1° bins (tuning curves smoothed with a circular Gaussian of s.d. 2°) and location into 2 cm bins (tuning maps smoothed with a Gaussian of s.d. 4 cm). We used 4-fold cross-validation over contiguous blocks of time: tuning curves were estimated from three folds and the held-out fold was decoded. The chance decoder always reports the most occupied state bin of the training folds. Errors are absolute angular differences for head direction and Euclidean distances for location.

Head direction cells We analyzed head direction cells from Duszkiewicz et al. [41]. Mice foraged for small food rewards in an open field: a modular arena (80×80 cm floor, 50 cm high walls) on a plastic platform inside a larger metal frame ($90 \times 90 \times 180$ cm), with a white cue card on one wall as the main distal landmark. Neural activity was recorded from the postsubiculum with 64-channel silicon probes. Units were classified as putative excitatory or fast-spiking inhibitory cells from waveform shape and mean firing rate, and excitatory cells whose head-direction information exceeded the 99th percentile of a time-reversed control distribution were labeled head direction cells. Position and head orientation were tracked at 100 Hz. We analyzed the 30 sessions of Datasets 1 and 2 of the release that were recorded in the square arena, and all cells labeled both excitatory and head-direction cells ($n = 1{,}489$).

Rate maps were computed as for the place cells, with 50 ms time bins, 20×20 location bins (4 cm), and 90 bins (4°) for head direction and for movement direction. For the location$\times$HD maps, time bins with speed below 2 cm/s and bins with fewer than 5 time bins were excluded; for the HD$\times$MD maps the thresholds were 1 cm/s and 10 time bins. Fold maps were smoothed with a Gaussian kernel of s.d. 1 bin along each axis (circular along angular axes). Movement direction was computed from the frame-to-frame displacement of the tracked position with the $\mathrm{atan2}$ function, wrapped to $[0^\circ, 360^\circ)$, smoothed with a circular boxcar filter of 20 samples (200 ms), and advanced by 3 samples (30 ms) relative to the position samples.

Decoding of location, HD and MD used the same decoder with $\tau = 50$ ms windows, excluding time bins with speed below 2 cm/s. Spike counts were smoothed in time with a Gaussian kernel (s.d. 100 ms) before decoding. Head direction and movement direction were binned into 1° bins with tuning curves smoothed by a circular Gaussian of s.d. 3°, and location into 32×32 bins (2.5 cm). The decoding analysis included the 30 sessions above and session A3708-200317b, 31 sessions in total.

Data and code availability

The place cell recordings from Pfeiffer and Foster [40] were provided directly by the authors. The head direction cell recordings are from Duszkiewicz et al. [41] and are publicly available. The analysis code (rate-map construction, the cross-fold correlation estimator, Bayesian decoding, and the deep network pipelines) is available at `https://github.com/mx60s/variance_analysis`.

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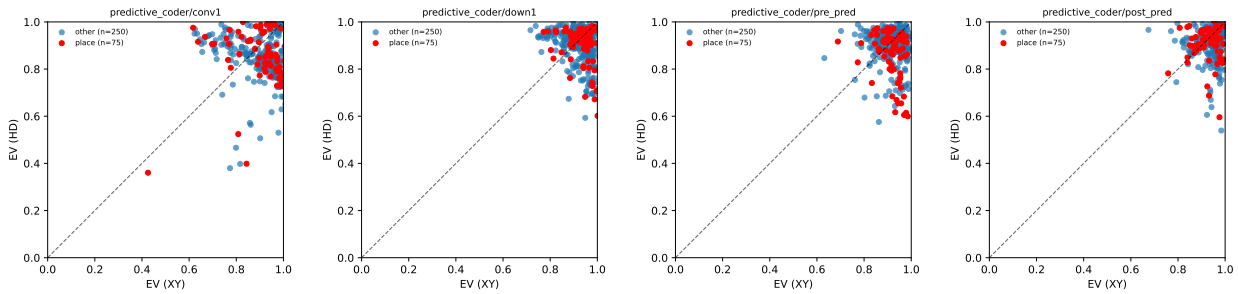
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Supplementary Information

Entangled location and head direction tuning across all layers of the deep networks

In the main text we quantified the contributions of 2-D location and head direction (HD) to the responses of model units for a subset of layers (Fig. 4j for the predictive coding network and Fig. 5 for ResNet-50). Here we report the same analysis for every layer we examined. For each unit we computed EV_{xy} and EV_{hd} (Eq. 2), with the two folds defined by camera pitch so that the two fold maps come from different images. Each point in the scatter plots below is one unit.

In every trained layer of both networks the population lies close to the diagonal $\mathrm{EV}_{xy} \approx \mathrm{EV}_{hd}$: location and head direction contribute comparably to the response of individual units, rather than each unit being selective to one variable and invariant to the other. This entanglement is the same in early and late layers, and for units classified as putative place cells and for the remaining units. It is therefore a general property of the spatial representations of these networks rather than a property of a particular layer. Biological place cells and head direction cells, in contrast, are each dominated by a single variable (Figs. 6, 7).



(a) conv1 (b) down1 (c) pre_pred (d) post_pred

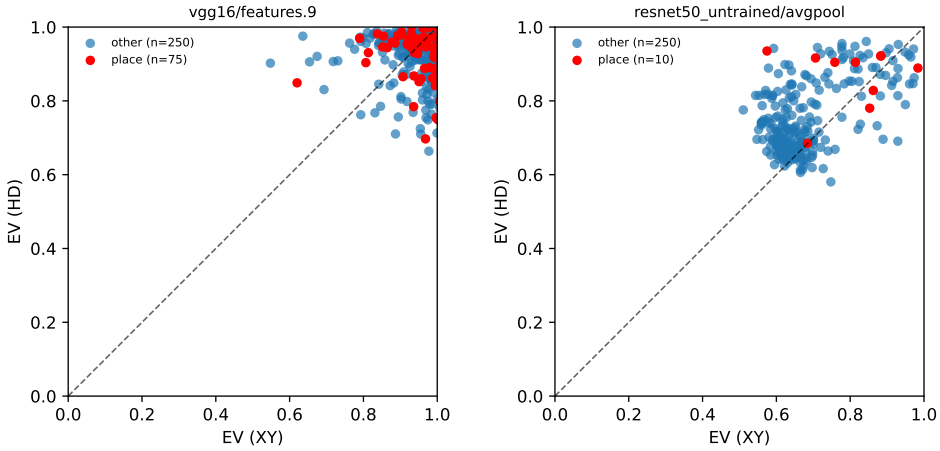
Figure S1: Location and head direction contribute comparably in all layers of the predictive coding network. Each panel shows the contribution of 2-D location (EV_{xy}, horizontal axis) versus head direction (EV_{hd}, vertical axis) for individual units in one layer; the diagonal marks $\mathrm{EV}_{xy} = \mathrm{EV}_{hd}$. In every layer the population clusters near the diagonal.

Scattered 3-D location$\times$HD firing fields in the CNN

For example ResNet-50 units that pass the 2-D place cell classification, we plot the full 3-D rate map over 2-D location and head direction. Unlike biological place cells, whose firing fields occupy the same location across head directions (Fig. 6a,b), these units fire only for particular combinations of location and head direction, so their fields are fragmented and scattered across head-direction slices. This is the CNN counterpart of the joint tuning shown for the predictive coding network in Fig. 4c–h.

The finite-sampling floor of the split-half estimator

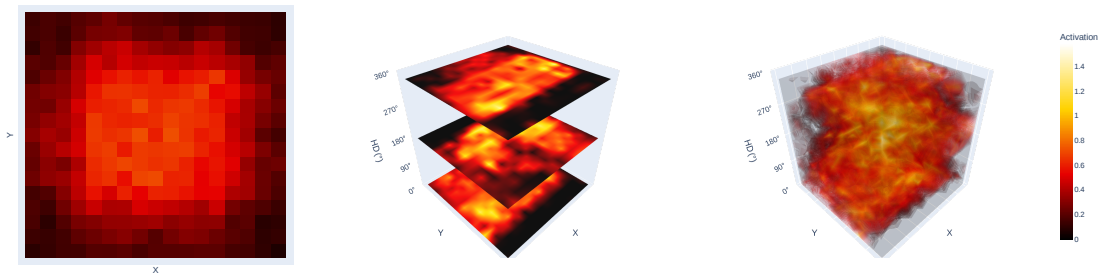
On an axis to which the neuron is invariant, the true EV is 0, but the estimate equals $1 - \langle C_{ii} \rangle_i$, because identical tuning curves estimated from two finite halves of the data correlate imperfectly. The number of samples per bin sets the reliability $\langle C_{ii} \rangle_i$ and therefore the floor. Fig. S4 shows the floor for the $\theta = 0°$ simulation of Fig. 3 as a



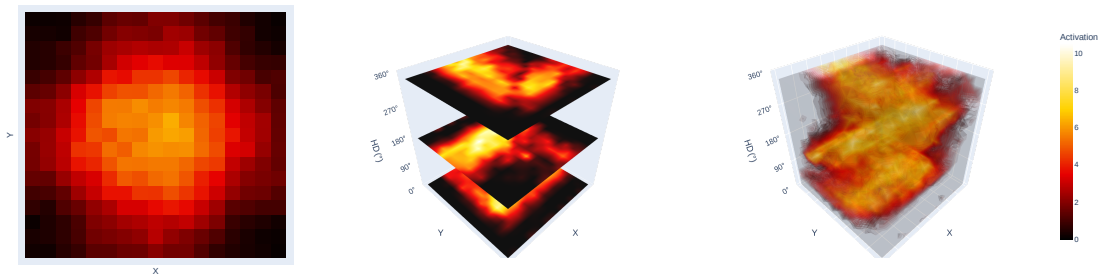
(a) VGG-16 `features.9` (b) untrained ResNet-50 `avgpool`

Figure S2: Location and head direction contribute comparably in other CNNs. As in Fig. S1, for (a) an early layer of an ImageNet-pretrained VGG-16 and (b) the `avgpool` layer of a randomly initialized ResNet-50 (Fig. 5 shows four layers of the pretrained ResNet-50). The VGG-16 layer lies on the $\mathrm{EV}_{xy} = \mathrm{EV}_{hd}$ diagonal like every trained layer we examined. The untrained network has lower EV for both variables, leans slightly towards head direction, and has 10 units that pass the place cell criteria.

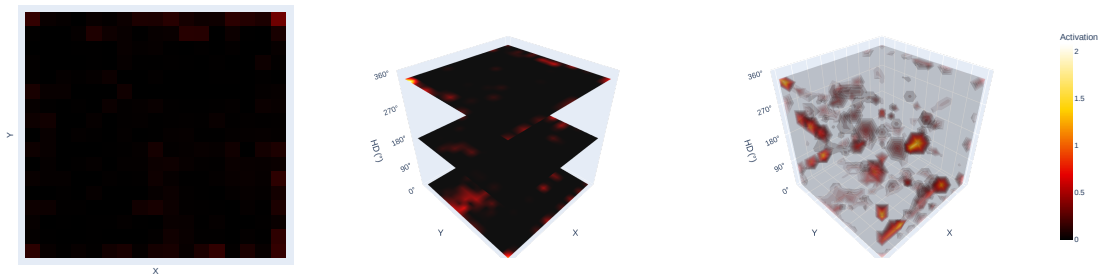
function of the number of bins per axis and the number of samples. With about 4 or more samples per bin, the floor is at most 0.05 and $\mathrm{EV}(y)$ equals it (15×15 bins at $N = 1{,}000$: $\mathrm{EV}(y) = 0.054$, floor 0.052). With 2 samples per bin or fewer, $\mathrm{EV}(y)$ exceeds the floor, and the reliability-normalized contribution no longer returns 0: at 30 bins per axis and $N = 1{,}000$ (1.1 samples per bin), $\mathrm{EV}(y) = 0.29$ with a floor of 0.14; at 15×15 bins and $N = 250$, $\mathrm{EV}(y) = 0.35$ with a floor of 0.23. The tuned axis is recovered in every condition ($\mathrm{EV}(x)$ between 0.99 and 1.01). We therefore report the floor together with EV for each dataset.



(a)



(b)



(c)

Figure S3: Firing fields of place-cell-like CNN units are scattered across the joint location–head-direction space. Each panel shows the 3-D rate map of one ResNet-50 unit classified as a putative place cell, as a function of 2-D location and head direction (HD slices). High activity is confined to particular location–HD combinations rather than to a single HD-invariant location, so the place field breaks up across HD. The near-diagonal contributions in Figs. 5 and S2 quantify this property.

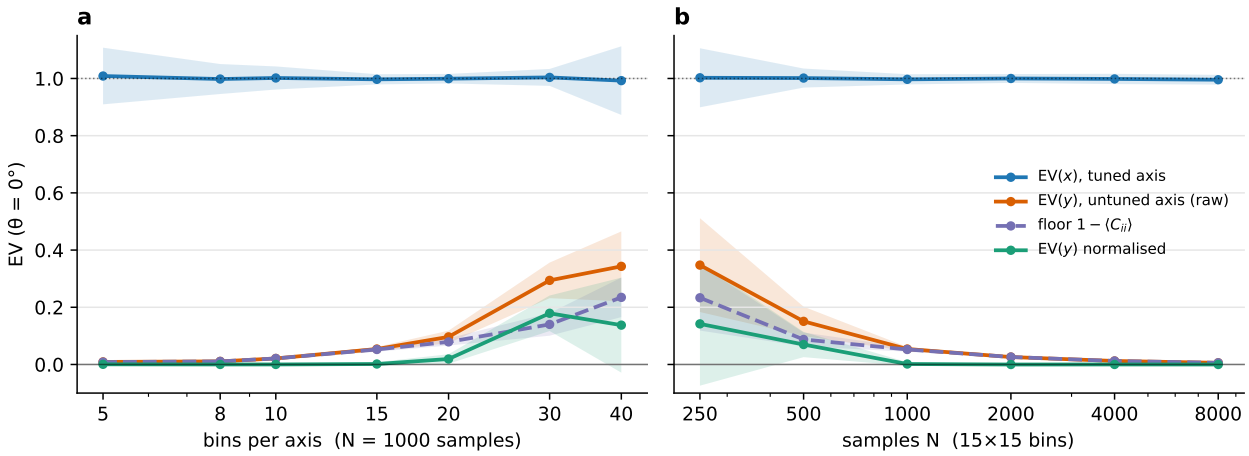


Figure S4: The finite-sampling floor of the split-half estimator is set by the reliability of the two folds. Simulated neuron tuned to x only ($\theta = 0°$), so the true $\mathrm{EV}(y) = 0$ and $\mathrm{EV}(x) = 1$. Lines are means and shaded bands ± 1 s.d. over 100 simulated data sets. (a) Varying the number of bins per axis at $N = 1{,}000$ samples. (b) Varying the number of samples at 15×15 bins. Blue: $\mathrm{EV}(x)$ on the tuned axis. Orange: $\mathrm{EV}(y) = 1 - \langle C_{ij} \rangle_{i \neq j}$ on the untuned axis. Dashed purple: the floor $1 - \langle C_{ii} \rangle_i$. Green: reliability-normalized $\mathrm{EV}_{\mathrm{norm}}(y)$. The untuned-axis estimate equals the floor down to about 4 samples per bin; with sparser sampling it exceeds the floor. The tuned axis is recovered in every condition.